Lecture 21 - Pathfinding Algorithms (Single Source)

Computer Systems, Data Structures, and Data Management (4CM508)

Dr Sam O'Neill

Pathfinding Demonstrations

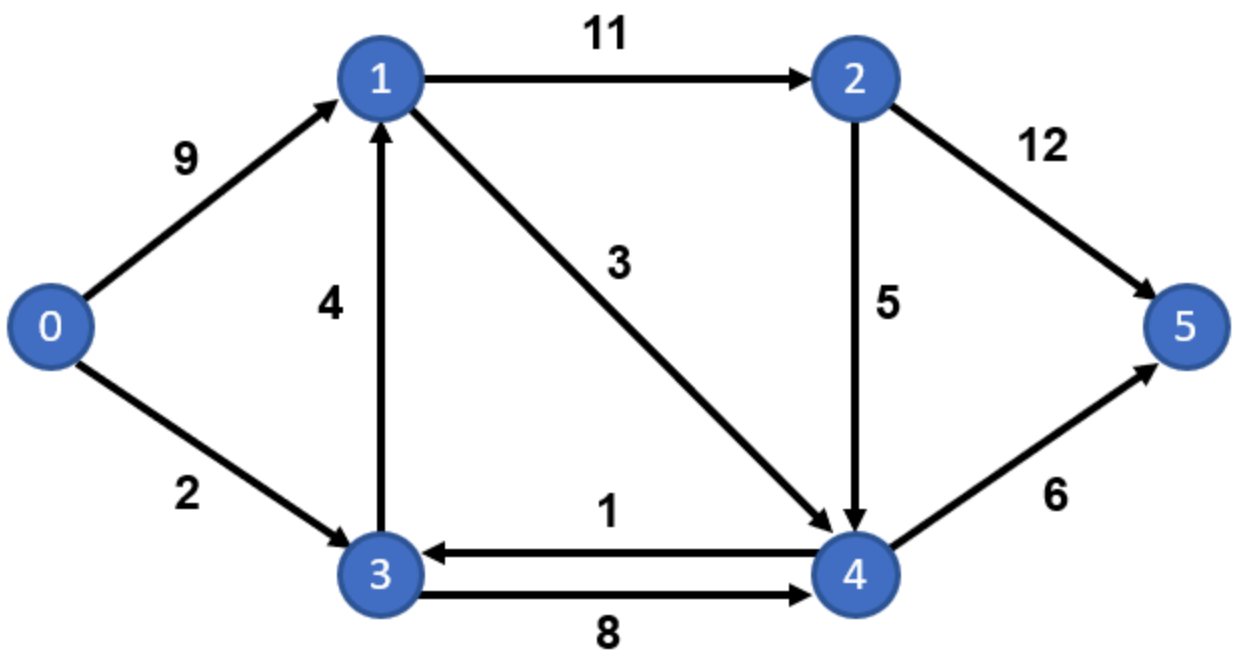
Interactive Pathfinding Playground

Visualalgo.net - Single-source Shortest Paths

Why?

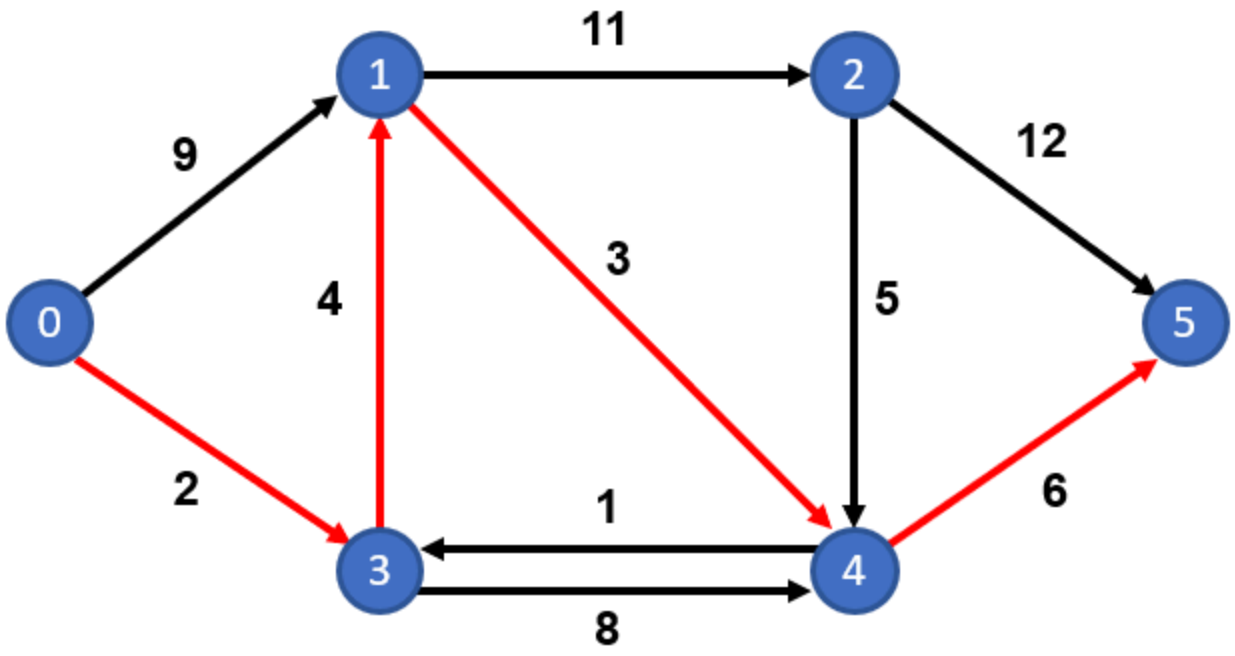
- GPS Navigation
- Video Games (NPCs)
- Robotics
- Logistics (Transportation of Goods)
- Network Routing

Blue Peter Example



What is the shortest path from 0 to 5?

Blue Peter Example



Shortest path between 0 and 5 is $0 - 3 - 1 - 4 - 5$ of length 15.

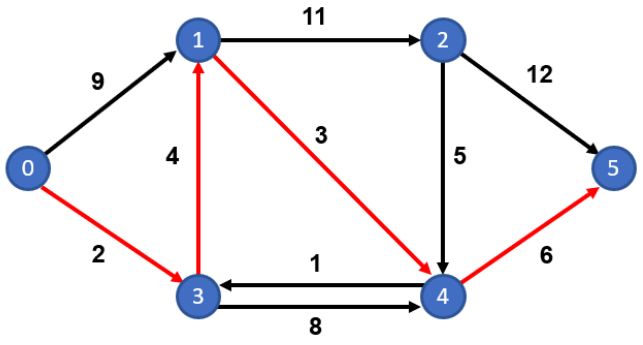
Paths

Walk

A walk is a sequence of vertices and edges of a graph.

Path

A path is a walk without repeated vertices. e.g. the edges in red are a path!

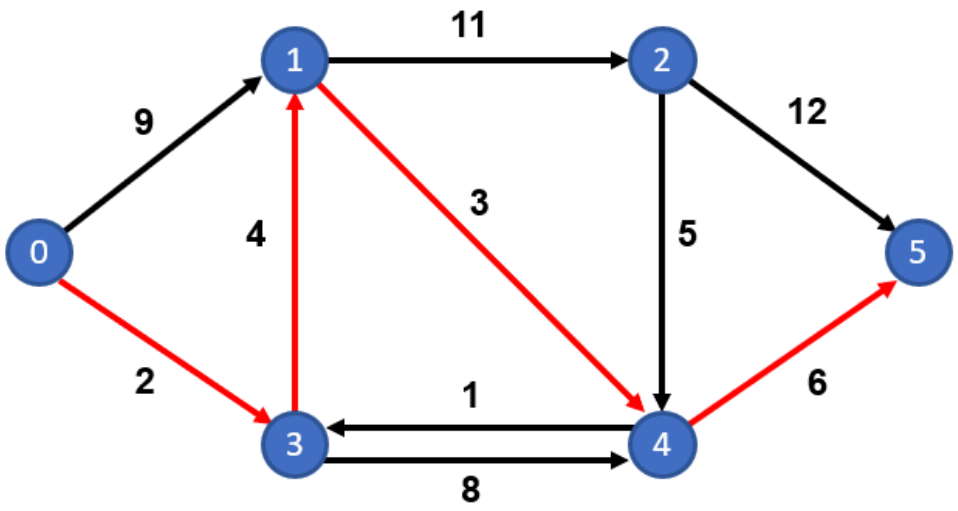


Single-source Shortest Path Problem

Find the shortest paths from a source s to **all other** vertices in the graph.

Single-pair Shortest Path Problem

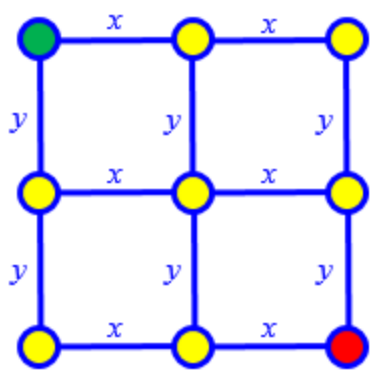
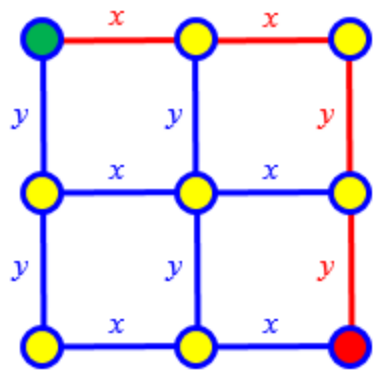
Find the shortest paths from a source s to a destination t in the graph.



Naive approach

- Search all the paths and work out there distance
- Pick the path with minimum distance

Number of Paths in a Grid (Corner to Corner)

4×4 Grid	Example Path - $xxyy$
	

You need 2 x's and 2 y's to get from s (green vertex) to t (red vertex).

Thus total number of paths is the number of ways you can write down 2 x's and 2 y's.

e.g. $xxyy$, $xyxy$, $xyyx$, $yxxy$, $yxyx$ and $yyxx$.

That is 6 paths.

Number of Paths From s to t

I won't dwell on the mathematics. If you are interested then look here.

However, in a 4×4 grid like the previous example there are $\binom{4}{2} = \frac{4!}{2!2!} = 6$ paths.

For an $n \times n$ grid there are $\binom{2n}{n} = \frac{(2n)!}{n!n!}$

Grid Size	No Paths
8×8	$12,870$
20×20	Approximately 1.38×10^{11} (one hundred and thirty billion)

These are very small. It is also much worse for a dense graph!

KEY POINT: It is impossible to search all paths!

Context

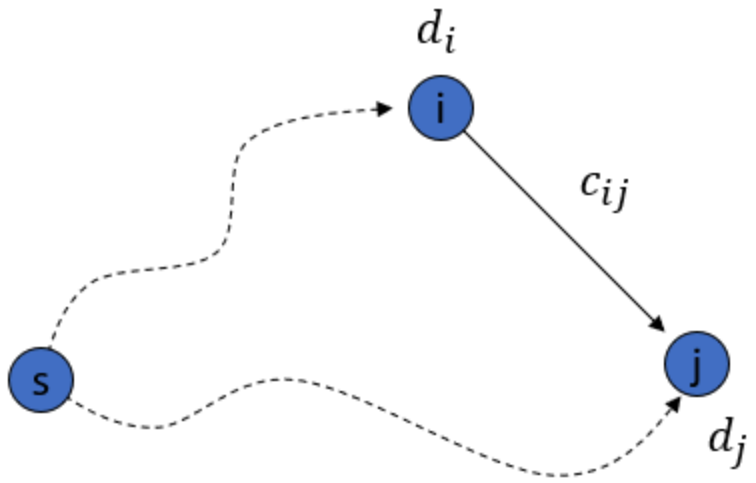
There are an estimated 10^{78} to 10^{82} atoms in the universe.

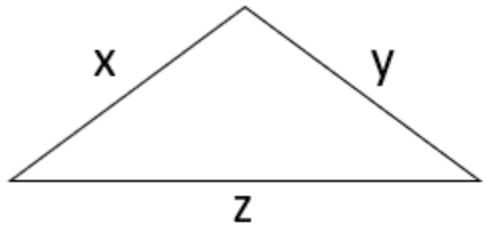
A 140×140 grid has approximately 10^{82}

That is a very very very very small graph! Oh dear!

Shortest Path Triangle Inequality

Imagine you know some path p_j from node s to node j that has a distance d_j





$x + y \geq z$

Then for that path p to be the shortest path:

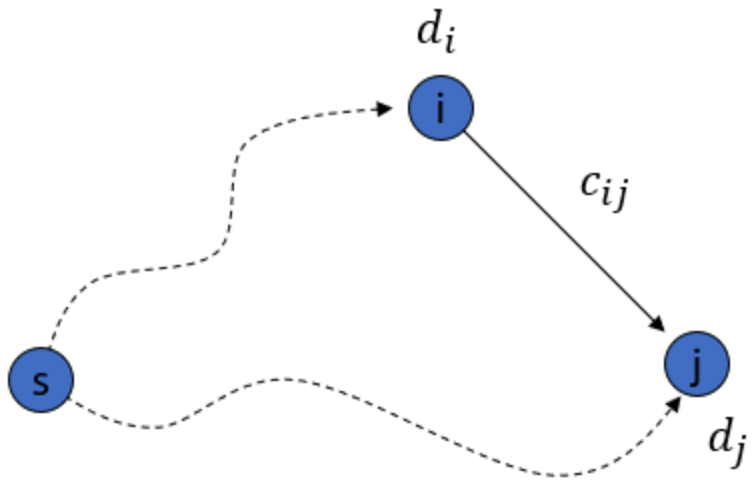
The distance from s to i + the cost of edge (i, j) **must be greater** than the current cost of reaching d_j for all edges (i, j).

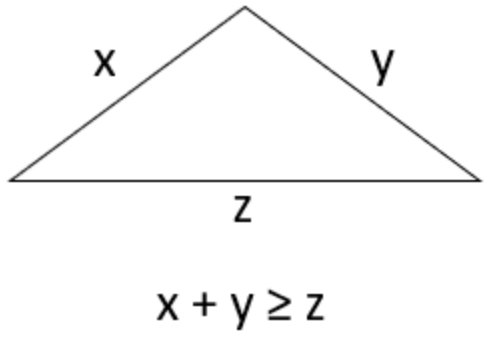
$$d_i + c_{ij} \geq d_j \quad \forall (i, j) \in E$$

Getting a Better Shortest Path

If for a path p_j from s to j, there exists a vertex i where:

$$d_i + c_{ij} < d_j$$





Then update the distance d_{ij} to path p_j to be:

$$d_j = d_i + c_{ij}$$

Dijkstra's Algorithm

Computes all the shortest paths from a source s to **all other** vertices in the graph.

1. Initialise the network, source vertex has a distance of 0 and all other vertices a distance of ∞, label all vertices as temporary
2. Label the temporary vertex with minimum distance as permanent
3. Update the distances and predecessors to reachable temporary vertices from the current permanent vertex
4. Repeat steps 2 and 3 until all vertices visited

This is a set of broad steps. We will see Python Code shortly.

Animations

Please see the animations on the module, it is the best way to learn what this algorithm is doing.

Using a Priority Queue

Having seen the animations, how do we track the distances and select the minimum each time?

With a priority queue!

- As you explore the graph, store the best know distance to vertex i.
- Select the vertex with minimum distance to be processed next.

Pseudocode

```
// https://en.wikipedia.org/wiki/Dijkstra%27s_algorithm
function Dijkstra(Graph, source):
    dist[source] ← 0                           // Initialization

    create vertex priority queue Q

    for each vertex v in Graph.Vertices:
        if v ≠ source
            dist[v] ← INFINITY                 // Unknown distance from source to v
            prev[v] ← UNDEFINED                // Predecessor of v

        Q.add_with_priority(v, dist[v])


    while Q is not empty:                      // The main loop
        u ← Q.extract_min()                    // Remove and return best vertex
        for each neighbor v of u:              // Go through all v neighbors of u
            alt ← dist[u] + Graph.Edges        // alternative distance(u, v)
            if alt < dist[v]:
                dist[v] ← alt
                prev[v] ← u
                Q.decrease_priority(v, alt)    // Update the distance of vertex v

    return dist, prev
```

Python Code

You can find Python code on the module

Proof of Optimality

The path found by the Dijkstra's will be the optimal shortest path.

We won't look at this, but you can read about it here.

Or just read any good book or article on the topic. e.g. - CLRS -Introduction to Algorithms

Time Complexity

Dijkstra's Algorithm has a worse case time complexity of:

- $O(|V|^2 \log(|V|))$ with an adjacency list
- $O((|V| + |E|) \log(|V|))$ with a binary heap
- Can get better, $O(|E| + |V| \log(|V|))$ with a Fibonacci heap

For a complete graph the number of edges $|E|$ is $O(|V|^2)$.

So efficient if the number of edges is small, otherwise the algorithm tends to be quadratic.

If you are interested see - CLRS -Introduction to Algorithms

Space Complexity

We maintain a list of permanent vertices $O(|V|)$ and min-priority queue of vertices $O(|V|)$ to be processed. Thus $O(|V|)$.

If you are interested see - CLRS -Introduction to Algorithms

A* Algorithm

If we are focusing on a single destination we can do better!

Idea

- Maintain an estimate for each vertex in the graph to the destination vertex t
- Update Dijkstra's algorithm to select the next vertex to process based on:
  Total Estimated Distance = Minimum Distance from s + Heuristic Distance to t
- Stop when we reach the goal

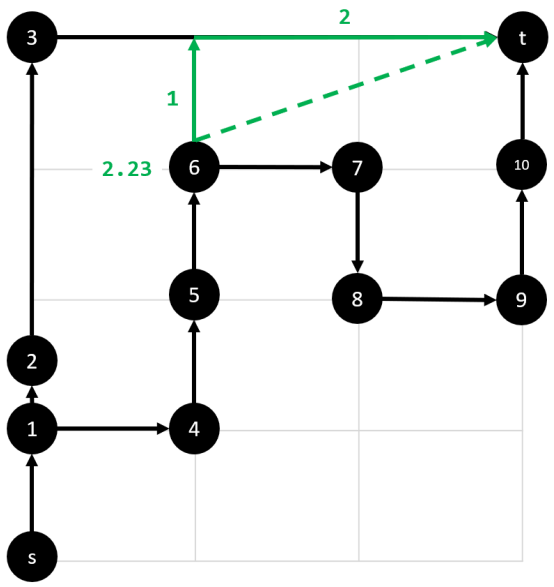
One way to think about this is Dijkstra's distance heuristic is just 0.

- So it selects based on the minimum distance from s (as before)

Example

Maintain the straight-line distance from each vertex to the destination vertex.

Here we measure the distance between vertices and the destination t. e.g. vertex 6 to t



You can think of it as seeking the destination.

Interactive Pathfinding Playground

Heuristic Function

The heuristic function is the estimate from vertex n to vertex t.

Total Estimated Distance = Shortest Distance from s + Heuristic Distance to t

$$f(n) = g(n) + h(n)$$

- n is vertex
- $g(n)$ is the **calculated** distance from source s to n
- $h(n)$ is the **estimated** distance from source n to t
- $f(n)$ is the total estimated distance

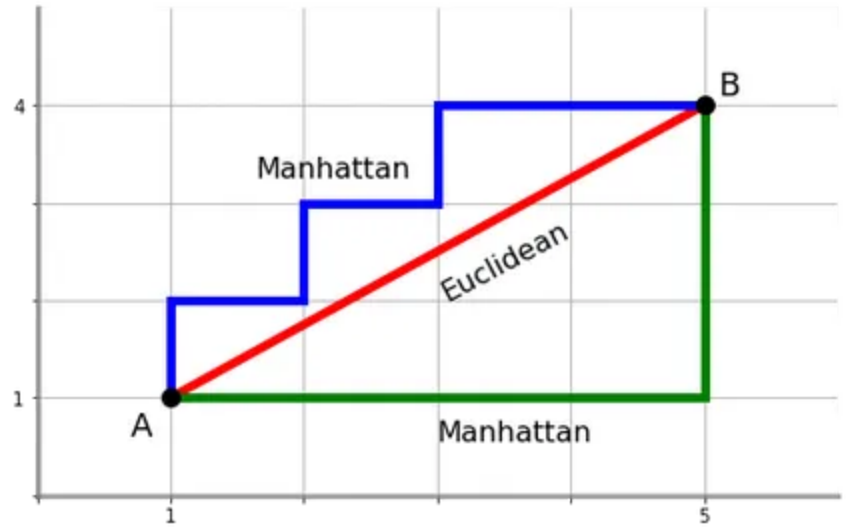
Euclidean and Manhattan Distance

Euclidean Distance

Here the estimated distance is the straight line distance.

Manhattan

Also known as the rectilinear and taxi-cab distance.



Worked example - Euclidean

https://www.101computing.net/a-star-search-algorithm/

Pseudocode

```
// This assumes heuristic(v) is consistent and admissable
function AStar(Graph, source, goal):
    dist[source] ← 0                            // Initialization

    create vertex priority queue Q

    for each vertex v in Graph.Vertices:
        if v ≠ source
            dist[v] ← INFINITY                  // Unknown distance from source to v
            prev[v] ← UNDEFINED                 // Predecessor of v

        Q.add_with_priority(v, dist[v])


    while Q is not empty:                       // The main loop
        u ← Q.extract_min()                     // Remove and return best vertex
        if u = goal:
            return dist, prev
        for each neighbor v of u:               // Go through all v neighbors of u
            alt ← dist[u] + Graph.Edges + heuristic(v)       // Estimated alternative distance(u, v)
            if alt < dist[v]:
                dist[v] ← alt
                prev[v] ← u
                Q.decrease_priority(v, alt)     // Update the distance of vertex v

    return None
```

Python Code

You can find Python code on the module.

Proof of Correctness

The path found by the A* algorithm will be the shortest path if the heuristic is:

- admissable
- consistent

Together these guarantee that **the heuristic never overestimates the true cost**.

Note there is more to this, but it summarised the main point

If you are interested you can look them up or read about it here.

Or just read any good book or article on the topic.

Time Complexity

This depends on the heuristic and is beyond the scope of this course.

It is enough to state that the worst-case is **exponential** if the heuristic is bad and the branching factor (average number of adjacent vertices a vertex has - i.e. its neighbours) is high.

With a good heuristic, it will have a worst-case time complexity similar to Dijkstra's, but on average outperforms it.

Summary

- Algorithms used in the real-world, every second of every day.
- Dijkstra's algorithm finds the shortest path.
  - Time Complexity
    - $O(|V|^2 \log(|V|))$ with an adjacency list
    - $O((|V| + |E|) \log(|V|))$ with a binary heap
    - Can get better, $O(|E| + |V| \log(|V|))$ with a Fibonacci heap
  - Space Complexity
    - $O(|V|)$
- A* algorithm also finds the shortest path
  - Uses a heuristic function to estimate the distance to the destination
  - Common heuristic functions are Euclidean and Manhattan distance

References

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C., 2022. Introduction to algorithms. MIT press.

Visualgo.net - Graphs

Interactive Pathfinding Playground

https://www.themathdoctors.org/how-many-paths-from-a-to-b/

http://theory.stanford.edu/~amitp/GameProgramming/AStarComparison.html